

Semantic Enrichment of the Laboratory Data Dictionary of the Study of Health in Pomerania (SHIP-START-4) with LOINC; Detailed Mapping Results

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1 Background and Significance

Insufficient semantic annotation can make it difficult to measure the comparability of laboratory test results in research [1]. This creates the need for data standardization and interoperability [1, 2]. The Logical Observation Identifiers Names and Codes (LOINC)¹ was developed by the Regenstrief Institute as a solution for sharing interoperable data in biomedicine [3]. LOINC serves as a comprehensive coding system for concepts related to medical laboratory science [4].

Unlike West Germany, high morbidity and mortality have been observed in East Germany over the last century [5, 6]. The regional population-based Study of Health in Pomerania (SHIP) therefore investigates the long-term progression of sub-clinical findings, their determinants and prognostic values, to acquire knowledge that facilitates early diagnosis and thus helps prevent the progression of disease [5, 7]. The SHIP covers various areas of patient health [8, 9, 10, 11, 12]. Each SHIP data set is accompanied by a data dictionary (DD), which provides descriptions of variables and definitions in an information database for the purposes of consistency, data validity, and reliability [13, 14]. The SHIP-START is a population-based cohort study conducted in northeastern Germany that consists of four measurement waves, the last of which serves as the basis for this study (SHIP-START 4) [7].

The FAIR (findability, accessibility, interoperability, reusability) data principles were conceived in 2014 by the Future of Research Communication and e-Scholarship (FORCE 11) community to promote better research data stewardship in the field of life sciences [15]. FAIRification is a gradual and iterative process that is contingent on close collaboration among stakeholders [16, 17]. A key step in the FAIRification process is the translation of human-readable data into machine-readable data in a manner that eliminates the need for specialized or ad hoc algorithms, translators, or mappings [18, 19].

In this work, we semantically enriched the laboratory data contained in the SHIP-START-4 laboratory DD, which was the most recently completed study follow-up (2019-2021) at the time of investigation. This work presents the detailed mapping results of our semantic enrichment process and provides detailed descriptions of the concepts applied in the process. The results of this work serve as a critical step towards improving its interoperability and hence FAIRness for the SHIP laboratory-related measurements.

2 Description of the SHIP-START-4 laboratory DD and file structure

The results of the semantic annotation of the DD are presented in tables. The following Table 1 describes the columns contained in the tables.

¹<https://loinc.org/>

Column	Description
var_shortn	Contains the abbreviation of the SHIP-START-4 code.
var_longn_eng	Contains the SHIP-START-4 code.
SHIP-START-4 Code Parts	Contains the component, property, time aspect, system, scale and method (abbreviated as C, P, T, S, Sc, M) of the selected SHIP-START-4 code.
LOINC	Contains the selected code from LOINC.
LOINC code parts	Contains the component, property, time aspect, system, scale and method (abbreviated as C, P, T, S, Sc, M) of the selected code from LOINC.
Relation (See Table 2)	Contains the description of the semantic relationship between the source and target codes.
Cardinality	Contains the description of the relationship between the source and target codes by referring to the number of matches found in LOINC for each SHIP-START-4 code.

Table 1. Description of the columns contained in the result tables

The following Table 2 describes the types of map relations, which is indicative of the semantic relationship between the source and target codes. This determines the degree of equivalence in their meaning and the overall suitability of the methodology used. The types of semantic relationships have been defined based on the W3C, SKOS Reference[19]².

Relation	Description	Symbol
Broader	The target of the map is broader in meaning than the source.	<
Equivalent	The source and target of the map have exactly the same meaning. This indicates that there is a high degree of confidence that the concepts can be used interchangeably in a wide range of information retrieval applications. In this work the following are considered to be equivalent: 1. Direct matches 2. Lexical variants 3. Spelling variants, including American or British 4. Synonyms 5. Semantically equivalent 6. Plurals can map to singular and vice versa	=
Inexact	The target mapping overlaps with the source code, but both source and target codes may cover additional meaning. Alternatively, the definitions are imprecise, and it is uncertain whether the source and target codes have the same boundaries to their meaning. In this case, a 'best fit' code shall be assigned [20]. For example, the source codes 'insulin/non-insulin dependent diabetes' shall be assigned codes that represent 'type 1/type 2 diabetes' in the target terminology as 'best fit'. If there is no 'best fit' code then the source code shall remain unmatched.	~
Narrower	The target of the map is narrower in meaning than the source.	>
No match/unmatched	There is no semantic match for the source code in the target code system.	≠

Table 2. Map-Relation Types

Details, such as the patient's age, sex and skin color are required when using the recommended formulae to calculate the eGFR. We assume that most of the SHIP participants are Caucasian because of the demographics at the time of sampling. Since 2021, new formulas for parameters used in the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) have been recommended. These were yet to be used by the related laboratory at the time of measurements used in this investigation. The results of the annotation of the SHIP-START-4 laboratory DD are contained in tables as follows:

- Table 3 shows the SHIP-START-4 medical laboratory DD codes that were not successfully mapped to LOINC codes.
- Table 4 shows the SHIP-START-4 medical laboratory DD codes that resulted in simple mappings when mapped to LOINC codes.

²<https://www.w3.org/TR/skos-reference/#L4858>

- Table 5 shows the SHIP-START-4 medical laboratory DD codes that resulted in complex mappings when mapped to LOINC codes.
- Table 6 shows the SHIP-START-4 medical laboratory DD codes that were successfully mapped to LOINC codes.

3 Annotation Results

The following Table 3 shows the SHIP-START-4 medical laboratory DD code —> LOINC mapping that was not successful.

Table 3: Unsuccessful Mappings

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
sg_u	LAB: (urine) density (kg/l)	C: Density P: kg/l T: Pt S: Urine Sc: Qn M:	—	—	≠	0

Table 3. SHIP-START-4 laboratory DD codes that were not successfully mapped to LOINC codes

The following Table 4 shows the simple SHIP-START-4 medical laboratory DD codes —> LOINC mappings.

Table 4: Simple Mappings

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
hgb	LAB: haemoglobin (mmol/l)	C: Haemoglobin P: mmol/l T: Pt S: Blood Sc: Qn M: Automated count	59260-0	C: Hemoglobin P: SCnc T: Pt S: Blood Sc: Qn M:	=	1
mch	LAB: MCH/mean corpuscular haemoglobin (fmol)	C: Erythrocyte mean corpuscular haemoglobin P: fmol T: Pt S: Erythrocytes Sc: Qn M: Automated count	59468-9	C: Erythrocyte mean corpuscular hemoglobin P: EntSub T: Pt S: Erythrocytes Sc: Qn M:	=	1
mchc	LAB: MCHC/mean corpuscular haemoglobin concentration (mmol/l)	C: Erythrocyte mean corpuscular haemoglobin concentration P: mmol/l T: Pt S: Erythrocytes Sc: Qn M: Automated count	59467-1	C: Erythrocyte mean corpuscular hemoglobin concentration P: SCnc T: Pt S: Erythrocytes Sc: Qn M:	=	1

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Table 4 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
normo_pct	LAB: normoblasts (%)	C: Normoblasts P: % T: Pt S: Blood Sc: Qn M:	33990-3	C: Normoblasts/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	1
normo	LAB: normoblasts (absolute) (Gpt/l)	C: Normoblasts (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	715-3	C: Normoblasts P: NCnc T: Pt S: Blood Sc: Qn M:	=	1
quick	LAB: QUICK/ thromboplastin time (%)	C: QUICK/ thromboplastin time P: % T: Pt S: PPP Sc: Qn M:	5894-1	C: Coagulation tissue factor induced actual/Normal P: RelTime T: Pt S: PPP Sc: Qn M:Coag	=	1
inr	LAB: thromboplastin time international normalized ratio	C: Thromboplastin time INR P: T: Pt S: PPP Sc: Qn M:	6301-6	C: Coagulation tissue factor induced.INR P: RelTime T: Pt S: PPP Sc: Qn M: Coag	=	1
ptt	LAB: PTT/ partial thrombo- plastin time (s)	C: Partial thromboplastin time P: s T: Pt S: PPP Sc: Qn M: Coag	14979-9	C: Coagulation surface induced P: Time T: Pt S: PPP Sc: Qn M: Coag	=	1
fib_cl	LAB: fibrinogen (Clauss) (g/l)	C: Fibrinogen (Clauss) P: g/l T: Pt S: PPP Sc: Qn M: Coag	3255-7	C: Fibrinogen P: MCnc T: Pt S: PPP Sc: Qn M: Coag	=	1
fib_p	LAB: fibrinogen (g/l)	C: Fibrinogen P: g/l T: Pt S: PPP Sc: Qn M: Coag.dervied	48664-7	C: Fibrinogen P: MCnc T: Pt S: PPP Sc: Qn M: Coag.dervied	=	1

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Table 4 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
cl_s	LAB: serum chloride (mmol/l)	C: Chloride P: mmol/l T: Pt S: Ser/Plas Sc: Qn M: Generic	2075-0	C: Chloride P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ca_s	LAB: serum calcium, total (mmol/l)	C: Calcium, total P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2000-8	C: Calcium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
na_s	LAB: Na/sodium (mmol/l)	C: Sodium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2951-2	C: Sodium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
k_s	LAB: K/potassium (mmol/l)	C: Potassium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2823-3	C: Potassium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
mg_s	LAB: magnesium (mmol/l)	C: Magnesium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	2601-3	C: Magnesium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
hba1c_ifcc	HbA1c (IFCC) (mmol/mol)	C: HbA1c P: mmol/mol T: Pt S: Blood Sc: Qn M: IFCC	59261-8	C: Hemoglobin A1c/ Hemoglobin. total P: SFr T: Pt S: Blood Sc: Qn M: IFCC	=	1
gluc_s	LAB: serum glucose (mmol/l)	C: Glucose P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14749-6	C: Glucose P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
crea_s	LAB: serum creatinine (μmol/l)	C: Creatinine P: μmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	14682-9	C: Creatinine P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1

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Table 4 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
hrs_s	LAB: serum uric acid (μmol/l)	C: Uric acid P: μmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	14933-6	C: Urate P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
cystatin	LAB: cystatin C (mg/l)	C: Cystatin C P: mg/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	33863-2	C: Cystatin C P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ck_s	LAB: creatine kinase (total) (μkatal/l)	C: Creatine kinase (total) P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	2157-6	C: Creatine kinase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ldh_s	LAB: lactate dehydrogenase (μkatal/l)	C: Lactate dehydrogenase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	14804-9	C: Lactate dehydrogenase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M: Reaction: lactate to pyruvate	=	1
amyl_s	LAB: serum amylase (alpha-amylase) (μkatal/l)	C: Amylase (alpha-amylase) P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1798-8	C: Amylase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ggt_s	LAB: gamma-glutamyl transferase (μkatal/l)	C: gamma-glutamyl transferase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	2324-2	C: Gamma glutamyl transferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
lip_s	LAB: lipase (μkatal/l)	C: Lipase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	3040-3	C: Triacylglycerol lipase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1

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Table 4 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
tg_s	LAB: triglycerides (total) (mmol/l)	C: Triglycerides (total) P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14927-8	C:Triglyceride P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
chol_s	LAB: cholesterol (total) (mmol/l)	C: Cholesterol (total) P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14647-2	C: Cholesterol P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
hdlch	LAB: HDL-cholesterol (mmol/l)	C: HDL - Cholesterol P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14646-4	C: Cholesterol.in HDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
crp_hs	LAB: CRP (high-sensitive) (mg/l)	C: CRP P: mg/l T: Pt S: Serum/Plasma Sc: Qn M: High-sensitive	30522-7	C: C reactive protein P: MCnc T: Pt S: Serum/Plasma Sc: Qn M: High sensitivity	=	1
ust_trueb	LAB: (urine) urine turbidity, clouding	C: Urine turbidity, clouding P: T: Pt S: Urine Sc: Ord M:	20627-6	C: Turbidity P: PrThr T: Pt S: Urine Sc: Ord M:	=	1
crea_u	LAB: (urine) creatinine (mmol/l)	C: Creatinine P: mmol/l T: Pt S: Urine Sc: Qn M:	14683-7	C: Creatinine P: SCnc T: Pt S: Urine Sc: Qn M:	=	1
alb_u	LAB: (urine) albumin (mg/l)	C: Albumin P: mg/l T: Pt S: Urine Sc: Qn M:	1754-1	C: Albumin P: MCnc T: Pt S: Urine Sc: Qn M:	=	1

Table 4. SHIP-START-4 laboratory DD codes that resulted in simple maps when mapped to LOINC codes

The following Table 5 shows the complex SHIP-START-4 medical laboratory DD codes —> LOINC mappings.

Table 5: Complex Mappings

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
wbc	LAB: WBC/leukocytes (Gpt/l)	C: Leucocytes P: Gpt/l T: Pt S: Blood Sc: Qn M:	6690-2	C: Leucocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26464-8	C: Leucocytes P: NCnc T: Pt S: Blood Sc: Qn M:	=	
rbc	LAB: RBC/erythrocytes (Tpt/l)	C: Erythrocytes P: Tpt/l T: Pt S: Blood Sc: Qn M:	789-8	C: Erythrocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26453-1	C: Erythrocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count"	=	
mcv	LAB: mcv/mean corpuscular volume (fl)	C: Erythrocyte mean corpuscular volume P: volume (fl) T: Pt S: Erythrocytes Sc: Qn M:	787-2	C: Erythrocyte mean corpuscular volume P: EntVol T: Pt S: RBC Sc: Qn M: Automated count	=	2
			30428-7	C: Erythrocyte mean corpuscular volume P: EntVol T: Pt S: RBC Sc: Qn M:	=	
rdw	LAB: RDW/red cell distribution width (%)	C: Red blood cell distribution width P: (%) T: Pt S: Erythrocytes Sc: Qn M:	788-0	C: Erythrocyte distribution width P: Ratio T: Pt S: RBC Sc: Qn M: Automated count	=	2

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			30385-9	C: Erythrocyte distribution width P: Ratio T: Pt S: RBC Sc: Qn M:	=	
plt	LAB: PLT/ thrombocytes (Gpt/l)	C: Platelets P: Gpt/l T: Pt S: Blood Sc: Qn M:	777-3	C: Platelets P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26515-7	C: Platelets P: NCnc T: Pt S: Blood Sc: Qn M:	=	
mpv	LAB: MPV/ mean platelet volume (fl)	C: Mean platelet volume P: fl T: Pt S: Blood Sc: Qn M:	32623-1	C: Platelet mean volume P: EntVol T: Pt S: Blood Sc: Qn M: Automated count	=	2
			28542-9	C: Platelet mean volume P: EntVol T: Pt S: Blood Sc: Qn M:	=	
ne_pct_e	LAB: neutrophils (%)	C: Neutrophils P:% T: Pt S: Blood Sc: Qn M:	770-8	C: Neutrophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26511-6	C: Neutrophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
ly_pct_e	LAB: lymphocytes (%)	C: Lymphocytes P: % T: Pt S: Blood Sc: Qn M:	736-9	C: Lymphocytes/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26478-8	C: Lymphocytes/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
mo_pct_e	LAB: monocytes (%)	C: Monocytes P: % T: Pt S: Blood Sc: Qn M:	5905-5	C: Monocytes/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26485-3	C: Monocytes/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
eo_pct_e	LAB: eosinophils (%)	C: Eosinophils P: % T: Pt S: Blood Sc: Qn M:	713-8	C: Eosinophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26450-7	C: Eosinophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
ba_pct_e	LAB: basophils (%)	C: Basophils P: % T: Pt S: Blood Sc: Blood M:	706-2	C: Basophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2

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var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			30180-4	C: Basophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
ne_e	LAB: neutrophils (absolute) (Gpt/l)	C: Neutrophils (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	751-8	C: Neutrophils P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26499-4	C: Neutrophils P: NCnc T: Pt S: Blood Sc: Qn M:	=	
ly_e	LAB: lymphocytes (absolute) (Gpt/l)	C: Lymphocytes (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	731-0	C: Lymphocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26474-7	C: Lymphocytes P: NCnc T: Pt S: Blood Sc: Qn M:	=	
gfr_larsson_t	LAB: GFR estimated (Larsson) (77.24* (cystatin ** -1.2623)) (ml/min)	C: GFR estimated (Larsson) P: ml/min T: Pt S: Serum/Plasma Sc: Qn M: (77.24* (cystatin **-1.2623)	50210-4	C: Glomerular filtration rate/ 1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Cystatin C - based formula	=	2
			69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M:	=	

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var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
ust_farb	LAB: (urine) urine color	C: Color P: T: Pt S: Urine Sc: Nom M:	50553-7	C: Color P: Type T: Pt S: Urine Sc: Nom M: Auto	=	2
			5778-6	C: Color P: Type T: Pt S: Urine Sc: Nom M:	=	
hct	LAB: haematocrit	C: Haematocrit P: T: Pt S: Blood Sc: Qn M:	4544-3	C: Hematocrit P: VFr T: Pt S: Blood Sc: Qn M: Automated count	=	3
			20570-8	C: Hematocrit P: VFr T: Pt S: Blood Sc: Qn M:	=	
			31100-1	C: Haematocrit P: VFr T: Pt S: Blood Sc: Qn M: Impedance	=	
hba1c	LAB: Hb A1c (%)	C: Hb A1c P: % T: Pt S: Blood Sc: Qn M:	4548-4	C: Hemoglobin A1c/Hemoglobin. total P: MFr T: Pt S: Blood Sc: Qn M:	=	2
			17856-6	C: Hemoglobin A1c/Hemoglobin. total P: MFr T: Pt S: Blood Sc: Qn M: HPLC	=	

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var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
gfrmdrd	LAB: GFR estimated (MDRD) (ml/min/1,73qm)	C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: MDRD	69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M:	=	3
		C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: MDRD	77147-7	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M: Creatinine-based formula (MDRD)	=	
		C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: MDRD	48642-3	C: Glomerular filtration rate/ 1.73 sq M.predicted. non black P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M: Creatinine-based formula (MDRD)	=	
gfrcapa	GFR (CAPA) (ml/min/1,73qm) calculated from cystatin	C: GFR P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: CAPA	69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M:	=	2

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
		C: GFR P: ml/min/1,73qm T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CAPA	50210-4	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Cystatin C- based formula	=	
gfr_ckdepi	Glomerular filtration rate (CKD-EPI for- mula) (ml/min) calculated from creatinine	C: Glomerular filtra- tion rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M:	=	3
		C: Glomerular filtra- tion rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	88294-4	C: Glomerular filtration rate/1.73 sq M. predicted. non black P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Creatinine- based formula (CKD-EPI)	=	
		C: Glomerular filtra- tion rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	62238-1	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Creatinine- based formula (CKD-EPI)	=	
alat_s	LAB: alanine aminotransferase (ALAT/GPT) (µkatal/l)	C: Alanine aminotransferase (ALAT/GPT) P: µkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1742-6	C: Alanine amino- transferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			1743-4	C: Alanine amino-transferase P: CCnc T: Pt S: Ser/Plas Sc: Qn M: With P-5'-P	=	
asat_s	LAB: aspartate aminotransferase (ASAT/GOT) (μkatal/l)	C: Aspartate aminotransferase (ASAT/GOT) P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1920-8	C: Aspartate aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			30239-8	C: Aspartate aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M: With P-5'-P	=	
alb_s	LAB: serum albumin (g/l)	C: Serum albumin P: g/l T: Pt S: Serum/Plasma Sc: Qn M:	1751-7	C: Albumin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			61152-5	C: Albumin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M: BCP	=	
ferri	Ferritin (μg/l)	C: Ferritin P: μg/l T: Pt S: Ser/Plas Sc: Qn M:	2276-4	C: Ferritin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			20567-4	C: Ferritin P: MCnc T: Pt S: Serum/Plasma S: Qn M: Immunoassay	=	
ldlch	LAB: LDL-cholesterol (mmol/l)	C: LDL-cholesterol P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	22748-8	C: Cholesterol.in LDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			69419-0	C: Cholesterol.in LDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Direct assay	=	
tsh	LAB: TSH/thyroid-stimulating hormone (mU/l)	C: TSH/thyroid-stimulating hormone P: mU/l T: Pt S: Serum/Plasma Sc: Qn M:	3016-3	C: Thyrotropin P: ACnc T: Pt S: Ser/Plas Sc: Qn M:	=	2
			11580-8	C: Thyrotropin P: ACnc T: Pt S: Serum/Plasma Sc: Qn M: Detection limit <= 0.005 mIU/L	=	
ft3	LAB: f T3/ triiodothyronine, free (pmol/l)	C: f T3/ triiodothyronine, free P: pmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14928-6	C: Triiodothyronine, free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			83126-3	C: Triiodothyronine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassay	=	
ft4	LAB: f T4/thyroxine, free (pmol/l)	C: f T4/thyroxine, free P: pmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14920-3	C: Thyroxine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			83121-4	C: Thyroxine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassay	=	

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
tpo_ak	LAB: TPO-AK/ anti-thyroid peroxidase antibody (U/ml)	C: TPO-AK/ anti-thyroid peroxidase antibody P: U/ml T: Pt S: Serum/Plasma Sc: Qn M:	8099-4	C: Thyroperoxi- dase Ab P: ACnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			56477-3	C: Thyroperoxi- dase Ab P: ACnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassy	=	
ph_u	LAB: (urine) pH	C: pH P: T: Pt S: Urine Sc: Qn M:	2756-5	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M:	=	3
			5803-2	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M: Test Strip	=	
			50560-2	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
leuco_u	LAB: (urine) leukocytes (Zellen/µl)	C: Leukocytes P: Zellen/µl T: Pt S: Urine Sc: Qn M:	30405-5	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M:	=	3
			20408-1	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			58805-3	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	

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Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
nitri_u	LAB: (urine) nitrite (pos/neg)	C: Nitrite P: pos/neg T: Pt S: Urine Sc: Ord M:	32710-6	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M:	=	3
			5802-4	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M: Test strip	=	
			50558-6	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M: Test strip. automated	=	
prot_u	LAB: (urine) protein (g/l)	C: Protein P: g/l T: Pt S: Urine Sc: Qn M:	2888-6	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M:	=	3
			5804-0	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			50561-0	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
gluc_u	LAB: (urine) glucose (mmol/l)	C: Glucose P: mmol/l T: Pt S: Urine Sc: Qn M:	15076-3	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			22705-8	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	

Continued on next page

Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			59156-0	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ket_u	LAB: (urine) ketone (mmol/l)	C: Ketones P: mmol/l T: Pt S: Urine Sc: Qn M:	22702-5	C: Ketones P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	2
			59158-6	C: Ketones P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ubg_u	LAB: (urine) urobilinogen (μ mol/l)	C: Urobilinogen P: μ mol/l T: Pt S: Urine Sc: Qn M:	34927-4	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			34928-2	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			60025-4	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
bili_u	LAB: (urine) bilirubin (μ mol/l)	C: Bilirubin P: μ mol/l T: Pt S: Urine Sc: Qn M:	70199-5	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			41016-7	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	

Continued on next page

Table 5 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			68367-2	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ery_u	LAB: (urine) erythrocytes (/µl)	C: Erythrocytes P: /µl T: Pt S: Urine Sc: Qn M:	30391-7	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M:	=	3
			20409-9	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			57747-8	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
hb_u	LAB: (urine) haemoglobin (mg/dl)	C: Hemoglobin P: mg/dl T: Pt S: Urine Sc: Qn M:	726-0	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: Qn M:	=	3
			49137-3	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			50559-4	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	

Table 5. SHIP-START-4 medical laboratory DD codes that resulted in complex maps when mapped to LOINC codes

The following Table 6 shows the successful SHIP-START-4 medical laboratory DD codes -> LOINC mappings.

Table 6: Successful Mappings

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
wbc	LAB: WBC/leukocytes (Gpt/l)	C: Leucocytes P: Gpt/l T: Pt S: Blood Sc: Qn M:	6690-2	C: Leucocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26464-8	C: Leucocytes P: NCnc T: Pt S: Blood Sc: Qn M:	=	
rbc	LAB: RBC/erythrocytes (Tpt/l)	C: Erythrocytes P: Tpt/l T: Pt S: Blood Sc: Qn M:	789-8	C: Erythrocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26453-1	C: Erythrocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	
hgb	LAB: haemoglobin (mmol/l)	C: Haemoglobin P: mmol/l T: Pt S: Blood Sc: Qn M:	59260-0	C: Hemoglobin P: SCnc T: Pt S: Blood Sc: Qn M:	=	1
mcv	LAB: mcv/mean corpuscular volume (fl)	C: Erythrocyte mean corpuscular volume P: volume (fl) T: Pt S: Erythrocytes Sc: Qn M:	787-2	C: Erythrocyte mean corpuscular volume P: EntVol T: Pt S: RBC Sc: Qn M: Automated count	=	2
			30428-7	C: Erythrocyte mean corpuscular volume P: EntVol T: Pt S: RBC Sc: Qn M:	=	

Continued on next page

Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
mch	LAB: MCH/mean corpuscular haemoglobin (fmol)	C: Erythrocyte mean corpuscular haemoglobin P: fmol T: Pt S: Erythrocytes Sc: Qn M:	59468-9	C: Erythrocyte mean corpuscular hemoglobin P: EntSub T: Pt S: Erythrocytes Sc: Qn M:	=	1
mchc	LAB: MCHC/mean corpuscular haemoglobin concentration (mmol/l)	C: Erythrocyte mean corpuscular haemoglobin concentration P: mmol/l T: Pt S: Erythrocytes Sc: Qn M:	59467-1	C: Erythrocyte mean corpuscular hemoglobin concentration P: SCnc T: Pt S: Erythrocytes Sc: Qn M:	=	1
rdw	LAB: RDW/red cell distribution width (%)	C: Red blood cell distribution width P: (%) T: Pt S: Erythrocytes Sc: Qn M:	788-0	C: Erythrocyte distribution width P: Ratio T: Pt S: RBC Sc: Qn M: Automated count	=	2
plt	LAB: PLT/thrombocytes (Gpt/l)	C: Platelets P: Gpt/l T: Pt S: Blood Sc: Qn M:	777-3	C: Platelets P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26515-7	C: Platelets P: NCnc T: Pt S: Blood Sc: Qn M:	=	
mpv	LAB: MPV/mean platelet volume (fl)	C: Mean platelet volume P: fl T: Pt S: Blood Sc: Qn M:	32623-1	C: Platelet mean volume P: EntVol T: Pt S: Blood Sc: Qn M: Automated count	=	2
			28542-9	C: Platelet mean volume P: EntVol T: Pt S: Blood Sc: Qn M:	=	

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
mpv	LAB: MPV/mean platelet volume (fl)	C: Mean platelet volume P: fl T: Pt S: Blood Sc: Qn M:	32623-1	C: Platelet mean volume P: EntVol T: Pt S: Blood Sc: Qn M: Automated count	=	2
ne_pct_e	LAB: neutrophils (%)	C: Neutrophils P: % T: Pt S: Blood Sc: Qn M:	770-8	C: Neutrophils/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26511-6	C: Neutrophils/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
mo_pct_e	LAB: monocytes (%)	C: Monocytes P: % T: Pt S: Blood Sc: Qn M:	5905-5	C: Monocytes/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26485-3	C: Monocytes/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
eo_pct_e	LAB: eosinophils (%)	C: Eosinophils P: % T: Pt S: Blood Sc: Qn M:	713-8	C: Eosinophils/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26450-7	C: Eosinophils/100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
ba_pct_e	LAB: basophils (%)	C: Basophils P: % T: Pt S: Blood Sc: Blood M:	706-2	C: Basophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M: Automated count	=	2
			30180-4	C: Basophils/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	
ne_e	LAB: neutrophils (absolute) (Gpt/l)	C: Neutrophils (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	751-8	C: Neutrophils P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26499-4	C: Neutrophils P: NCnc T: Pt S: Blood Sc: Qn M:	=	
ly_e	LAB: lymphocytes (absolute) (Gpt/l)	C: Lymphocytes (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	731-0	C: Lymphocytes P: NCnc T: Pt S: Blood Sc: Qn M: Automated count	=	2
			26474-7	C: Lymphocytes P: NCnc T: Pt S: Blood Sc: Qn M:	=	
normo_pct	LAB: normoblasts (%)	C: Normoblasts P: % T: Pt S: Blood Sc: Qn M:	33990-3	C: Normoblasts/ 100 leukocytes P: NFr T: Pt S: Blood Sc: Qn M:	=	1
normo	LAB: normoblasts (absolute) (Gpt/l)	C: Normoblasts (absolute) P: Gpt/l T: Pt S: Blood Sc: Qn M:	715-3	C: Normoblasts P: NCnc T: Pt S: Blood Sc: Qn M:	=	1

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
quick	LAB: QUICK/ thromboplastin time (%)	C: QUICK/ thromboplastin time P: % T: Pt S: PPP Sc: Qn M:	5894-1	C: Coagulation tissue factor induced actual/Normal P: RelTime T: Pt S: PPP Sc: Qn M: Coag	=	1
inr	LAB: throm- boplastin time international normalized ratio	C: Thromboplastin time INR P: T: Pt S: PPP Sc: Qn M:	6301-6	C: Coagulation tissue factor induced.INR P: RelTime T: Pt S: PPP Sc: Qn M: Coag	=	1
ptt	LAB: PTT/partial thromboplastin time (s)	C: Partial thromboplastin time P: s T: Pt S: PPP Sc: Qn M: Coag	14979-9	C: Coagulation surface induced P: Time T: Pt S: PPP Sc: Qn M: Coag	=	1
fib_cl	LAB: fibrinogen (Clauss) (g/l)	C: Fibrinogen (Clauss) P: g/l T: Pt S: PPP Sc: Qn M: Coag	3255-7	C: Fibrinogen P: MCnc T: Pt S: PPP Sc: Qn M: Coag	=	1
fib_p	LAB: fibrinogen (g/l)	C: Fibrinogen P: g/l T: Pt S: PPP Sc: Qn M: Coag.dervied	48664-7	C: Fibrinogen P: MCnc T: Pt S: PPP Sc: Qn M: Coag.dervied	=	1
cl_s	LAB: serum chloride (mmol/l)	C: Chloride P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	2075-0	C: Chloride P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ca_s	LAB: serum calcium, total (mmol/l)	C: Calcium, total P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2000-8	C: Calcium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
na_s	LAB: Na/sodium (mmol/l)	C: Sodium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2951-2	C: Sodium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
k_s	LAB: K/potassium (mmol/l)	C: Potassium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	2823-3	C: Potassium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
mg_s	LAB: magnesium (mmol/l)	C: Magnesium P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	2601-3	C: Magnesium P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
hba1c_ifcc	HbA1c (IFCC) (mmol/mol)	C: HbA1c P: mmol/mol T: Pt S: Blood Sc: Qn M: IFCC	59261-8	C: Hemoglobin A1c/ Hemoglobin. total P: SFr T: Pt S: Blood Sc: Qn M: IFCC	=	1
gluc_s	LAB: serum glucose (mmol/l)	C: Glucose P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14749-6	C: Glucose P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
crea_s	LAB: serum creatinine (μmol/l)	C: Creatinine P: μmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	14682-9	C: Creatinine P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
hrs_s	LAB: serum uric acid (μmol/l)	C: Uric acid P: μmol/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	14933-6	C: Urate P: SCnc T: Pt S: Serum/Plasma S: Qn M:	=	1
cystatin	LAB: cystatin C (mg/l)	C: cystatin C P: mg/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	33863-2	C: Cystatin C P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
gfr_larsson_t	LAB: GFR estimated (Larsson) (77.24* (cystatin**-1.2623)) (ml/min)	C: GFR estimated (Larsson) P: ml/min T: Pt S: Serum/Plasma Sc: Qn M: (77.24* (cystatin**-1.2623))	50210-4	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Ser/Plas/Bld S: Qn M: Cystatin C-based formula	=	2

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Ser/Plas/Bld S: Qn M:	=	
ck_s	LAB: creatine kinase (total) (μkatal/l)	C: Creatine kinase (total) P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	2157-6	C: Creatine kinase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ldh_s	LAB: lactate dehydrogenase (μkatal/l)	C: Lactate dehydrogenase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	14804-9	C: Lactate dehydrogenase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M: Reaction: lactate to pyruvate	=	1
amyl_s	LAB: serum amylase (alpha-amylase) (μkatal/l)	C: Amylase (alpha-amylase) P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1798-8	C: Amylase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
ggt_s	LAB: gamma-glutamyltransferase (μkatal/l)	C: gamma-glutamyltransferase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M: Generic	2324-2	C: Gamma glutamyl transferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
lip_s	LAB: lipase (μkatal/l)	C: Lipase P: μkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	3040-3	C: Triacylglycerol lipase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
tg_s	LAB: triglycerides (total) (mmol/l)	C: Triglycerides (total) P: mmol/l T: Pt S: Serum/Plasma Sc: Qn	14927-8	C: Triglyceride P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
chol_s	LAB: cholesterol (total) (mmol/l)	C: Cholesterol (total) P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14647-2	C: Cholesterol P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
hdlch	LAB: HDL-cholesterol (mmol/l)	C: HDL - Cholesterol P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14646-4	C: Cholesterol.in HDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	1
crp_hs	CRP (high-sensitive) (mg/l)	C: CRP P: mg/l T: Pt S: Serum/Plasma Sc: Qn M: High-sensitive	30522-7	C: C reactive protein P: MCnc T: Pt S: Serum/Plasma Sc: Qn M: High sensitivity	=	1
ust_trueb	LAB: (urine) urine turbidity, clouding	C: urine turbidity, clouding P: T: Pt S: Urine Sc: Ord M:	20627-6	C: Turbidity P: PrThr T: Pt S: Urine Sc: Ord M:	=	1
ust_farb	LAB: (urine) urine color	C: Color P: T: Pt S: Urine Sc: Nom M:	50553-7	C: Color P: Type T: Pt S: Urine Sc: Nom M: Auto	=	2
			5778-6	C: Color P: Type T: Pt S: Urine Sc: Nom M:	=	
crea_u	LAB: (urine) creatinine (mmol/l)	C: Creatinine P: mmol/l T: Pt S: Urine Sc: Qn M:	14683-7	C: Creatinine P: SCnc T: Pt S: Urine Sc: Qn M:	=	1
hct	LAB: haematocrit	C: Haematocrit P: T: Pt S: Blood Sc: Qn M:	4544-3	C: Hematocrit P: VFr T: Pt S: Blood Sc: Qn M: Automated count	=	3

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			20570-8	C: Hematocrit P: VFr T: Pt S: Blood Sc: Qn M:	=	
			31100-1	C: Haematocrit P: VFr T: Pt S: Blood Sc: Qn M: Impedance	=	
hba1c	LAB: Hb A1c (%)	C: Hb A1c P: % T: Pt S: Blood Sc: Qn M:	4548-4	C: Hemoglobin A1c/Hemoglobin. total P: MFr T: Pt S: Blood Sc: Qn M:	=	2
			17856-6	C: Hemoglobin A1c/Hemoglobin. total P: MFr T: Pt S: Blood Sc: Qn M: HPLC	=	
gfrmdrd	LAB: GFR estimated (MDRD) (ml/min/1,73qm)	C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: MDRD	69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M:	=	3
		C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/Blood Sc: Qn M: MDRD	77147-7	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/Blood Sc: Qn M: Creatinine-based formula (MDRD)	=	

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
		C: GFR estimated P: ml/min/1,73qm T: Pt S: Serum/Plasma/ Blood Sc: Qn M: MDRD	48642-3	C: Glomerular filtration rate/ 1.73 sq M. predicted.non black P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Creatinine- based formula (MDRD)	=	
gfrcapa	GFR (CAPA) (ml/min/1,73 qm)	C: GFR P: ml/min/1,73qm T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CAPA	69405-9	C: Glomerular filtration rate/ 1.73 sq M. predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M:	=	2
		C: GFR P: ml/min/1,73qm T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CAPA	50210-4	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Cystatin C- based formula	=	
gfr_ckdepi	Glomerular filtration rate (CKD-EPI formula) (ml/min)	C: Glomerular filtration rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	69405-9	C: Glomerular filtration rate/1.73 sq M.predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M:	=	3

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
		C: Glomerular filtration rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	88294-4	C: Glomerular filtration rate /1.73 sq M. predicted.non black P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Creatinine-based formula (CKD-EPI)	=	
		C: Glomerular filtration rate P: ml/min T: Pt S: Serum/Plasma/ Blood Sc: Qn M: CKD-EPI formula	62238-1	C: Glomerular filtration rate/ 1.73 sq M. predicted P: ArVRat T: Pt S: Serum/Plasma/ Blood Sc: Qn M: Creatinine-based formula (CKD-EPI)	=	
alat_s	LAB: alanine aminotransferase (ALAT/GPT) (µkatal/l)	C: Alanine aminotransferase (ALAT/GPT) P: µkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1742-6	C: Alanine aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
		C: Alanine aminotransferase (ALAT/GPT) P: µkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1743-4	C: Alanine aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M: With P-5'-P	=	
asat_s	LAB: aspartate aminotransferase (ASAT/GOT) (µkatal/l)	C: Aspartate aminotransferase (ASAT/GOT) P: µkatal/l T: Pt S: Serum/Plasma Sc: Qn M:	1920-8	C: Aspartate aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
		C: Aspartate aminotransferase (ASAT/GOT) P: μ katal/l T: Pt S: Serum/Plasma Sc: Qn M:	30239-8	C: Aspartate aminotransferase P: CCnc T: Pt S: Serum/Plasma Sc: Qn M: With P-5'-P	=	
alb_s	LAB: serum albumin (g/l)	C: Serum albumin P: g/l T: Pt S: Serum/Plasma Sc: Qn M:	1751-7	C: Albumin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			61152-5	C: Albumin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M: BCP	=	
ferri	LAB: Ferritin (μ g/l)	C: Ferritin P: μ g/l T: Pt S: Serum/Plasma Sc: Qn M:	2276-4	C: Ferritin P: MCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			20567-4	C: Ferritin P: MCnc T: Pt S: Serum/Plasma S: Qn M: Immunoassay	=	
ldlch	LAB: LDL-cholesterol (mmol/l)	C: LDL-cholesterol P: mmol/l T: Pt S: Serum/Plasma Sc: Qn M:	22748-8	C: Cholesterol.in LDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			69419-0	C: Cholesterol.in LDL P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Direct assay	=	
tsh	LAB: TSH/thyroid-stimulating hormone (mU/l)	C: TSH/thyroid-stimulating hormone P: mU/l T: Pt S: Serum/Plasma Sc: Qn M:	3016-3	C: Thyrotropin P: ACnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			11580-8	C: Thyrotropin P: ACnc T: Pt S: Serum/Plasma Sc: Qn M: Detection limit ≤ 0.005 mIU/L	=	
ft3	LAB: f T3/ triiodothyronine, free (pmol/l)	C: f T3/ triiodothyronine, free P: pmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14928-6	C: Triiodothyro- nine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			83126-3	C: Triiodothyro- nine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassay	=	
ft4	LAB: fT4 /thyroxine, free (pmol/l)	C: f T4/thyroxine, free P: pmol/l T: Pt S: Serum/Plasma Sc: Qn M:	14920-3	C: Thyroxine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			83121-4	C: Thyroxine.free P: SCnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassay	=	
tpo_ak	LAB: TPO- AK/ anti- thyroid peroxidase antibody (U/ml)	C: TPO-AK/ anti-thyroid peroxidase antibody P: U/ml T: Pt S: Serum/Plasma Sc: Qn M:	8099-4	C: Thyroperoxi- dase Ab P: ACnc T: Pt S: Serum/Plasma Sc: Qn M:	=	2
			56477-3	C: Thyroperoxi- dase Ab P: ACnc T: Pt S: Serum/Plasma Sc: Qn M: Immunoassay	=	

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
ph_u	LAB: (urine) pH	C: pH P: T: Pt S: Urine Sc: Qn M: urine test strip	2756-5	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M:	=	3
			5803-2	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M: Test Strip	=	
			50560-2	C: pH P: LsCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
leuco_u	LAB: (urine) leukocytes (Zellen/µl)	C: Leukocytes P: Zellen/µl T: Pt S: Urine Sc: Qn M: urine test strip	30405-5	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M:	=	3
			20408-1	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			58805-3	C: Leukocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
nitri_u	LAB: (urine) nitrite (pos/neg)	C: Nitrite P: pos/neg T: Pt S: Urine Sc: Ord M: urine test strip	32710-6	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M:	=	3
			5802-4	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M: Test strip	=	

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			50558-6	C: Nitrite P: PrThr T: Pt S: Urine Sc: Ord M: Test strip. automated	=	
prot_u	LAB: (urine) protein (g/l)	C: Protein P: g/l T: Pt S: Urine Sc: Qn M: Urine test strip	2888-6	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M:	=	3
			5804-0	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			50561-0	C: Protein P: MCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
gluc_u	LAB: (urine) glucose (mmol/l)	C: Glucose P: mmol/l T: Pt S: Urine Sc: Qn M: urine test strip	15076-3	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			22705-8	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			59156-0	C: Glucose P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ket_u	LAB: (urine) ketone (mmol/l)	C: Ketones P: mmol/l T: Pt S: Urine Sc: Qn M: Urine test strip	22702-5	C: Ketones P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	2

Continued on next page

Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			59158-6	C: Ketones P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ubg_u	LAB: (urine) urobilinogen ($\mu\text{mol/l}$)	C: Urobilinogen P: $\mu\text{mol/l}$ T: Pt S: Urine Sc: Qn M: urine test strip	34927-4	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			34928-2	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			60025-4	C: Urobilinogen P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
bili_u	LAB: (urine) bilirubin ($\mu\text{mol/l}$)	C: Bilirubin P: $\mu\text{mol/l}$ T: Pt S: Urine Sc: Qn M: urine test strip	70199-5	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M:	=	3
			41016-7	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			68367-2	C: Bilirubin P: SCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
ery_u	LAB: (urine) erythrocytes (/ μl)	C: Erythrocytes P: / μl T: Pt S: Urine Sc: Qn M: urine test strip	30391-7	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M:	=	3

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Table 6 – continued from previous page

var_shortn	var_longn_eng	SHIP-START-4 Code Parts	LOINC	LOINC code parts	Relation	Cardinality
			20409-9	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			57747-8	C: Erythrocytes P: NCnc T: Pt S: Urine Sc: Qn M: Test strip. automated	=	
hb_u	LAB: (urine) haemoglobin (mg/dl)	C: Hemoglobin P: mg/dl T: Pt S: Urine Sc: Qn M:	726-0	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: Qn M: urine test strip	=	3
			49137-3	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: Qn M: Test strip	=	
			50559-4	C: Hemoglobin P: MCnc T: Pt S: Urine Sc: SemiQn M: Test strip. automated	=	
alb_u	LAB: (urine) albumin (mg/l)	C: Albumin P: mg/l T: Pt S: Urine Sc: Qn M:	1754-1	C: Albumin P: MCnc T: Pt S: Urine Sc: Qn M:	=	1

Table 6. SHIP-START-4 medical laboratory DD codes that were successfully mapped to LOINC codes

4 Sharing/Access information

The DD was derived on 28.05.2024 from:

<https://transfer.ship-med.uni-greifswald.de/FAIRequest/data-use-explore>

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